

Institute of Actuaries of India

Subject CS2A – Risk Modelling and Survival Analysis (Paper A)

March 2021 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

- i) Answer – (d) [2]
- ii) Answer – (b) [4]
- iii) Answer – (a) [2]
- iv) B = 0.00015, c = 1.1000; however, the option (c) shows the values in different order.
i. Answer – Full marks for either of the option (c) OR (e). [2]
- v) Answer – (b) [2]
- vi) Answer – (a) [2]
- vii) Answer – (b) [4]
- viii) Answer – (b) [2]
- ix) Answer – (b) [4]
- x) Answer – (a) [5]

[29 Marks]**Solution 2:**

- i) We have that s_1 , the state of the system in period 1, is given by $s_1 = [0.55, 0.45]$ with $s_2 = [0.67, 0.33]$ and $s_3 = [0.70, 0.30]$

Assuming the researcher is correct then s_1 and s_2 are linked by $s_2 = s_1 (P)$ where P is the transition matrix as to how buyer changes buying preference. We also have that s_3 and s_2 are linked by $s_3 = s_2 (P)$.

Now we have that P will be a 2 by 2 matrix, and that the elements of each row of P add to one, so that we can write

$$P = \begin{bmatrix} x_1 & 1 - x_1 \\ x_2 & 1 - x_2 \end{bmatrix} \quad [1]$$

Where x_1 and x_2 are unknown.

Using $s_2 = s_1(P)$ we have

$$\begin{aligned} [0.67, 0.33] &= [0.55, 0.45] \begin{bmatrix} x_1 & 1 - x_1 \\ x_2 & 1 - x_2 \end{bmatrix} \\ [0.70, 0.30] &= [0.67, 0.33] \begin{bmatrix} x_1 & 1 - x_1 \\ x_2 & 1 - x_2 \end{bmatrix} \end{aligned} \quad [1]$$

Hence, expanding, we have

$$0.67 = 0.55x_1 + 0.45x_2 \quad (1)$$

$$0.33 = 0.55(1 - x_1) + 0.45(1 - x_2) \quad (2)$$

$$0.70 = 0.67x_1 + 0.33x_2 \quad (3)$$

$$0.30 = 0.67(1 - x_1) + 0.33(1 - x_2) \quad (4) \quad [1]$$

Equation (2), when rearranged, becomes equation (1) and similarly equation (4), when rearranged, becomes equation (3). Hence we have two simultaneous equations (equations (1) and (3)) in two unknowns.

From equation (1)

$$x_1 = (0.67 - 0.45x_2)/0.55$$

so substituting for x_1 in equation (3) we get

$$0.70 = 0.67[(0.67 - 0.45x_2)/0.55] + 0.33x_2$$

$$\text{i.e. } (0.70)(0.55) = (0.67)(0.67) - (0.67)(0.45)x_2 + (0.33)(0.55)x_2$$

$$\text{i.e. } x_2 = [(0.67)(0.67) - (0.70)(0.55)] / [(0.67)(0.45) - (0.33)(0.55)]$$

$$\text{i.e. } x_2 = 0.5325 \text{ and}$$

$$x_1 = (0.67 - 0.45x_2)/0.55 = 0.7825$$

[2]

Hence our estimate of the transition matrix P is equal to

$$P = \begin{vmatrix} 0.7825 & 0.2175 \\ 0.5325 & 0.4675 \end{vmatrix}$$

[1]

[6]

ii) The market shares for period 4 are given by

$$s_4 = s_3(P)$$

$$[0.70, 0.30] \begin{vmatrix} 0.7825 & 0.2175 \\ 0.5325 & 0.4675 \end{vmatrix}$$

$$[0.7075, 0.2925]$$

$$\text{i.e. } s_4 = [0.7075, 0.2925]$$

and note here that the elements of s_4 add to one (as required).

Hence the estimated market shares of two products in period 4 are 70.75% and 29.25%.

[2]

[8 Marks]

Solution 3:

i) Graduation refers to the process of using statistical techniques to improve the estimates provided by the crude rates. The aims of graduation are to produce a smooth set of rates that are suitable for a particular purpose, to remove random sampling errors (as far as possible) and to use the information available from adjacent ages to improve the reliability of the estimates. Graduation results in a 'smoothing' of the crude rates.

[2]

ii) Graduation cannot remove any bias in the data arising from faulty data collection or otherwise.

[1]

iii) Smoothness versus adherence to data: Smoothness and adherence to data are usually conflicting requirements. Perfect smoothness pays little or no attention to the data, while perfect adherence to the data means no smoothing at all. If the graduation process results in rates that are smooth but show little adherence to the data, then the rates are over-graduated. Over-graduation has an opposite, referred to as under-graduation. This refers to the case where insufficient smoothing has been carried out. This will tend to produce a curve of inadequate smoothness, but better adherence to data.

[1.5]

Suitability for the purpose in hand: the suitability of a graduation for practical work depends very much on what that work is, and can only be assessed in particular cases. However, two very important observations are:

a) In life insurance work, losses result from premature deaths (benefits are paid sooner than expected) so we must not underestimate mortality

b) In pensions or annuity work, losses result from delayed deaths (benefits are paid for longer than expected) so we must not overestimate mortality.

[1.5]

[3]

[6 Marks]

Solution 4:

- i) A false positive outcome is undesirable because the patient may be caused unnecessary worry or required to undergo further tests or treatment before it is established that they do not actually have the disease. A false negative outcome is also undesirable because the patient may not now be identified early enough to receive effective treatment for the disease.

Since Covid-19 is an infectious disease, there is the additional concern that false negative patients may unknowingly spread the disease to other people.

[2]

ii)

Actual \ Predicted(→)	YES	NO	TOTAL
YES	TP=30	FN=0	30
NO	FP=0	TN=170	170
Total	30	170	200

[2]

iii) The four measures are

- Precision = $TP/(FP+TP) = 30/(0+30) = 100\%$
- Recall = $TP/(TP+FN) = 30/(30+0) = 100\%$
- F_1 Score = $2 \times \text{precision} \times \text{recall} / (\text{precision} + \text{recall}) = 2 \times 100\% \times 100\% / (100\% + 100\%) = 100\%$
- False Positive rate = $FP/(TN+FP) = 0/(170+0) = 0\%$

[0.5 each = 2]

iv) Let T_i be the time until default of bond i where $i = 1, 2, 3$. We want to calculate the joint probability:

$$P(T_1 \leq 1, T_2 \leq 1, T_3 \leq 1) = C[u_1, u_2, u_3]$$

$$\text{Where } u_i = P[T_i \leq 1] = 0.1 \text{ for } i = 1, 2, 3$$

[1]

Using the Gumbel copula with parameter $\alpha = 2$, we have:

$$\begin{aligned} P(T_1 \leq 1, T_2 \leq 1, T_3 \leq 1) &= c[u_1, u_2, 3] \\ &= \text{Exp}\{-(-\ln u_1)^2 + (-\ln u_2)^2 + (-\ln u_3)^2\}^{1/2} \\ &= 0.0185 \text{ or } 1.85\% \end{aligned}$$

[2]

[1]

[4]

v)

- The Gumbel copula exhibits (non-zero) upper-tail dependence, the degree of which can be varied by adjusting the single parameter. However, it exhibits no lower tail dependence.
- Hence, the Gumbel copula is appropriate if we believe that the three investments are likely to behave similarly as the term approaches five years but not at early durations.
- This is unlikely to be the case though. If one bond defaults early on, then it may be indicative of problems in the industry sector or the economy and so the other investments may also be likely to default early on.
- If we believe the performance of investments issued by companies within the industry are much more closely associated (eg subject to the same systemic and operational risk factors), then a copula that exhibits both lower and upper tail dependence, such as the Student's t copula, may be more appropriate.

[1 each = 4]

[14 Marks]**Solution 5:**

The number of annual claims N follows a binomial distribution:

$$N \sim B(100, 0.4) \text{ then}$$

$$E(N) = 100 \times 0.4 = 40$$

and

$$\text{Var}(N) = 100 \cdot 0.4 \cdot 0.6 = 24.$$

[1]

Let X denote the distribution of the individual claim amounts, so that $X \sim \text{Pareto}(10, 9,000)$.

Then

$$\begin{aligned} E(X) &= 9000/(10-1) \\ &= 1,000 \end{aligned}$$

and

$$\begin{aligned} \text{Var}(X) &= (9000^2 \times 10)/(9^2 \times 8) \\ &= 1,250,000. \end{aligned}$$

[1]

The annual aggregate claim amount S has

$$E(S) = E(N)E(X) = 40 \cdot 1,000 = 40,000$$

[1]

and

$$\begin{aligned} \text{Var}(S) &= E(N)\text{Var}(X) + \text{Var}(N)E(X)^2 \\ &= 40 \cdot 1,250,000 + 24 \cdot 1,000^2 \\ &= 74,000,000 \\ &= (8,602.33)^2 \end{aligned}$$

[1]

[4 Marks]

Solution 6:

i) The characteristic equation is given by

$$1 - (8/15)\lambda + (1/15)\lambda^2 = \{1 - (1/3)\lambda\}X\{1 - (1/5)\lambda\} = 0$$

which has roots = 3 and 5. They are both greater than 1. Hence, subject to the initial values having appropriate distributions, this implies (weak) stationarity.

[2]

ii) Note that $\text{Cov}(X_t, Z_t) = 1$ and $\text{Cov}(X_t, Z_{t-1}) = 8/15 - 1/7 = 41/105$

We need to generate 3 distinct equations linking γ_0 , γ_1 and γ_2 . This can be done as follows:

(A)

$$\begin{aligned} \gamma_0 &= \text{Cov}(X_t, X_t) = \text{Cov}(1 + (8/15)X_{t-1} - (1/15)X_{t-2} + Z_t - (1/7)Z_{t-1}, X_t) \\ &= 8/15 \gamma_1 - 1/15 \gamma_2 + 1 - 1/7 \times 41/105 \\ &= 8/15 \gamma_1 - 1/15 \gamma_2 + 694/735 \end{aligned}$$

[1]

(B)

$$\begin{aligned} \gamma_1 &= \text{Cov}(X_t, X_{t-1}) = \text{Cov}(1 + (8/15)X_{t-1} - (1/15)X_{t-2} + Z_t - (1/7)Z_{t-1}, X_{t-1}) \\ &= 8/15 \gamma_0 - 1/15 \gamma_1 - 1/7 \end{aligned}$$

[1]

(C)

$$\begin{aligned} \gamma_2 &= \text{Cov}(X_t, X_{t-2}) = \text{Cov}(1 + (8/15)X_{t-1} - (1/15)X_{t-2} + Z_t - (1/7)Z_{t-1}, X_{t-2}) \\ &= 8/15 \gamma_1 - 1/15 \gamma_0 \end{aligned}$$

[1]

Solving these equations, we get,

$$\gamma_0 = 1.1753$$

$$\gamma_1 = 0.4537$$

$$\gamma_2 = 0.1636$$

[2]

There fore

$$\rho_0 = 1$$

$$\rho_1 = \gamma_1 / \gamma_0 = 0.386$$

$$\rho_2 = \gamma_2 / \gamma_0 = 0.139$$

[1]

Now, $P_k = (8/15) P_{k-1} - (1/15) P_{k-2}$ for $k \geq 2$

[1]

It is to be shown that the solution has the form

$$P_k = A(1/3)^k + B(1/5)^k$$

Substituting the proposed solution above into the recurrence relationships gives

$$\begin{aligned} (8/15) P_{k-1} - (1/15) P_{k-2} &= (8/15)[A(1/3)^{k-1} + B(1/5)^{k-1}] - (1/15)[A(1/3)^{k-2} + B(1/5)^{k-2}] \\ &= A(1/3)^k ((8/15) \times 3 - (1/15) \times 9) + B(1/5)^k ((8/15) \times 5 - (1/15) \times 25) \\ &= A(1/3)^k + B(1/5)^k \\ &= P_k \end{aligned}$$

[3]

So the solution does have this form.

The values of A and B are fixed by $P_0 = 1$ and $P_1 = 0.386$

Since $A + B = 1$

$$(1/3)A + (1/5)B = 0.386$$

[2]

Solving above two equations

$$A = 1.395$$

$$B = -0.395$$

So,

$$P_k = 1.395(1/3)^k - 0.395(1/5)^k$$

[1]

[13]

[15 Marks]

Solution 7:

i) The three parameters of GEV are

- A location parameter α
- A scale parameter β , this is always greater than zero
- A shape parameter γ

[2]

ii) The first two parameters rescale the distribution and determine shift and stretch of the distribution. The third parameter determines overall shape of the distribution and its sign.

[2]

iii) These are analogous to following in the standard distribution

- Mean
- Standard deviation
- Skewness

[2]

[6 Marks]

Solution 8:

i) We can re-write the journal data for manufacturing defects of mobile phones, using terminology for mortality investigations as follows:

t	Defects	Censored	Lives
0			100
1		7	93
2	1	2	90
3			90
4		2	88
5	2	3	83
6			83
7			83
8	1	1	81
9			81
10			81
11	2		79
12			79

(Note that even though no calls were made on 30/06, it was confirmed on 31/07 that there were no manufacturing defects for $t=6$ and $t=7$) [2]

Assuming that phone sets which were censored at any time t were at risk of defects at time t , then we can calculate the required statistics as follows, where t is the time since purchase of a new phone, calculated in months:

J	t_j	N_j	d_j	λ_j	$\Lambda(t)$	$\hat{S}(t)$
0	0	100			0.00000	1.00000
1	2	93	1	0.01075	0.01075	0.98930
2	5	88	2	0.02273	0.03348	0.96707
3	8	83	1	0.01205	0.04553	0.95549
4	11	81	2	0.02469	0.07022	0.93219

Where,

t_j is the j th time of defect

n_j is the number of phones available for defect at each time of defect

d_j is the number of defects at each time of defect

$\lambda_j = d_j/n_j$ is the estimates of the discrete hazard rates at each time of defect

$\Lambda(t)$ is the estimate of cumulative hazard function

$\hat{S}(t) = \exp(-\Lambda(t))$ is the estimate of the survival function

[3]

From the above, we can see that $\hat{S}(t > 11) = 0.9322$

Therefore, the probability of getting defective in one year is estimated to be $1 - 0.9322 = 6.78\%$.

[2]

[7]

ii)

From part (i), the probability of getting defective in one year = 6.78%

Therefore, the expected cost of providing the warranty is equal to:

$6.78\% \times 10,000 \text{ phones} \times \text{Rs}25,000 = \text{Rs}1,69,52,707 \sim \text{Rs}1.7 \text{ crores}$

[1]

An approximate 95% confidence interval for the integrand hazard function is:

$$\hat{\Lambda}_t \pm 1.96\sqrt{\text{Var}(\hat{\Lambda}_t)}$$

Where,

$$\text{Var}[\hat{\Lambda}_t] \approx \sum_{t_j \leq t} (d_j(n_j - d_j) / n_j^3)$$

[1]

We can calculate this as follows:

J	t_j	N_j	d_j	$\text{Var}(\hat{\Lambda}_t)$	$\sum t_j \leq t$
0	0	100	0	0	0
1	2	93	1	0.000114	0.000114
2	5	88	2	0.000252	0.000367

3	8	83	1	0.000143	0.000510
4	11	81	2	0.000297	0.000807

[2]

Therefore, an approximate 95% confidence interval for integrand hazard function for $t > 12$ is: $0.07022 \pm 1.96 \times 0.000807 = (0.1259, 0.0145)$ [1]

And an approximate 95% confidence interval for the survival function for $t > 12$ is equal to $(0.8817, 0.9856)$

...implying an approximate 95% confidence interval for probability of getting defective in 12 months equal to $(11.83\%, 1.44\%)$ [1]

...and an approximate 95% confidence interval for the estimated cost of providing replacement warranty equal to $(Rs2,95,77,660 ; Rs36,04,645)$ [1]

[7]

iii) The finance director's crude estimate does not allow for data that has been lost to the investigation. For instance, we do not have any information about the phones purchased by the seven individuals who provided incorrect contact details. Similarly, we do not have subsequent information in respect of those who requested not to be disturbed in the future or changed their contact details. It is possible that phones purchased by these individuals may have also developed manufacturing defects within the first year of purchase. Therefore, the crude rates not allowing for censoring introduces a downward bias in the finance director's estimates. Nelson-Aalen estimate allows for censored data by calculating estimates of the discrete hazard rates at each time of death as number of deaths divided by is the number of lives available to die at each time of death. The denominator in this calculation is reduced by the number of lives censored by the time of death. Therefore, the estimate of probability of defects in the first year will be higher using this approach (since it allows for the possibility of censored lives dying as well).

[4]

[18 Marks]
